

ARLINGTON, TX, USA

WMO: 722479

Lat: 32.664N

Lon: 97.094W

Elev: 192

StdP: 99.04

Time Zone: -6.00 (NAC)

Period: 99-19

WBAN: 53907

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.6	-2.5	-12.8	1.3	1.0	-10.9	1.5	3.4	11.8	9.5	10.9	11.3	3.5	340	0.495

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	11.5	38.4	23.2	37.3	23.4	36.0	23.4	25.6	32.7	25.2	32.3	24.8	31.8	4.2	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.9	19.2	27.8	23.3	18.5	27.5	22.9	18.0	27.4	79.4	32.8	77.5	32.6	75.8	31.8	27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
10.7	9.3	8.4	-8.4	40.0	2.1	1.9	-9.9	41.4	-11.1	42.5	-12.3	43.6	-13.9	45.0		
			-9.3	26.4	1.9	0.6	-10.7	26.9	-11.8	27.2	-12.9	27.6	-14.3	28.0		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	19.4	8.4	10.4	14.8	19.1	23.4	27.6	29.4	29.8	25.9	19.9	14.1	9.2
	DBStd	8.72	5.43	5.97	5.12	4.10	3.40	2.25	2.06	2.47	3.37	4.64	5.24	5.24
	HDD10.0	276	96	62	16	1	0	0	0	0	0	1	21	79
	HDD18.3	1185	311	229	130	39	6	0	0	0	1	38	145	287
	CDD10.0	3698	44	72	167	275	414	528	601	614	478	308	143	54
	CDD18.3	1565	2	6	22	62	162	278	342	356	229	86	17	3
	CDH23.3	18267	17	44	154	480	1491	3251	4524	4824	2550	799	112	21
	CDH26.7	9106	1	6	24	111	537	1601	2519	2777	1233	276	18	2
Wind	WSAvg	4.1	4.1	4.5	4.7	4.9	4.6	4.2	3.6	3.3	3.2	3.7	3.9	4.0
Precipitation	PrecAvg	959	59	63	79	95	109	100	59	58	79	101	73	70
	PrecMax	1399	117	174	149	187	212	281	154	147	263	249	186	242
	PrecMin	483	5	0	0	36	10	18	0	0	5	0	12	6
	PrecStd	233	32	49	42	43	52	81	43	35	70	71	53	58
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.5	27.5	29.0	31.8	34.5	37.4	39.7	40.6	38.2	33.8	28.9	25.8
		MCWB	15.0	16.0	18.0	20.9	23.1	23.4	23.1	22.6	22.4	21.7	18.9	17.6
	2%	DB	22.7	24.5	27.1	29.3	32.7	36.0	38.0	38.8	36.0	31.9	26.3	22.8
		MCWB	14.6	16.0	17.9	20.0	22.7	23.4	23.4	23.2	22.6	21.7	18.1	16.1
	5%	DB	20.4	22.3	25.0	27.8	31.2	34.6	36.9	37.6	34.2	29.6	24.1	20.6
		MCWB	14.1	15.4	17.0	19.7	22.4	23.4	23.6	23.5	22.4	20.3	17.3	15.0
	10%	DB	17.9	20.1	23.0	26.2	29.7	33.4	35.5	36.3	32.8	27.8	22.4	18.1
		MCWB	12.7	15.0	16.4	18.8	21.8	23.4	23.7	23.5	22.3	19.5	17.0	13.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.1	19.9	20.8	23.3	25.2	26.3	26.1	26.0	25.5	24.2	21.5	19.7
		MCDB	21.9	23.6	25.9	28.6	31.6	33.1	32.9	33.4	31.4	29.0	24.8	22.9
	2%	WB	17.6	18.8	19.7	22.1	24.2	25.4	25.5	25.3	24.7	23.4	20.2	18.3
		MCDB	20.3	21.9	24.3	27.1	29.7	32.1	32.8	33.1	30.6	28.2	23.6	21.5
	5%	WB	15.7	17.5	18.9	21.2	23.6	24.9	25.1	25.0	24.2	22.6	19.1	16.5
		MCDB	18.9	20.6	23.1	25.6	29.0	31.4	32.5	32.7	29.9	27.3	22.8	19.1
	10%	WB	12.9	15.7	17.9	20.4	22.9	24.3	24.7	24.5	23.7	21.5	17.9	14.3
		MCDB	17.1	19.2	21.9	24.5	28.2	30.6	31.9	32.1	29.4	26.0	21.5	17.7
Mean Daily Temperature Range	5% DB	MDBR	11.8	11.4	11.5	11.5	10.6	10.4	11.3	11.5	11.5	12.0	11.7	11.3
		MCDBR	15.5	14.2	13.2	12.6	11.9	11.9	13.0	13.1	12.8	13.4	13.0	14.3
		MCWBR	9.3	8.4	6.8	6.2	4.7	3.2	3.2	3.2	3.6	5.7	6.9	8.8
	5% WB	MCDBR	12.7	11.9	10.9	9.9	10.1	9.8	11.0	11.2	10.4	10.5	10.5	11.0
		MCWBR	9.5	8.7	6.9	5.9	4.8	3.4	3.3	3.2	3.6	5.0	7.2	9.0
Clear-Sky Solar Irradiance	taub	0.314	0.328	0.357	0.395	0.425	0.436	0.439	0.439	0.400	0.359	0.338	0.319	
	taud	2.502	2.470	2.406	2.323	2.299	2.301	2.331	2.314	2.371	2.472	2.472	2.489	
	Ebn at Noon	908	928	921	895	866	852	847	844	865	881	871	880	
	Edh at Noon	86	98	113	128	132	132	127	128	116	97	88	83	
All-Sky Solar Radiation	RadAvg	2.90	3.40	4.48	5.44	5.86	6.59	6.72	6.16	5.18	4.07	3.07	2.56	
	RadStd	0.29	0.54	0.35	0.43	0.54	0.62	0.46	0.40	0.43	0.60	0.39	0.33	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (48 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page